

Acquired heart disease in children part I

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Rheumatic fever/ Rheumatic heart disease

Definition

:- inflammatory disease that can involve the heart, joints, skin, and brain.

:-Typically develops 2-4weeks after a streptococcal throat infection.

Epidemiology

- 0.3% - GAS tonsillitis treated with Penicillin.
- 3% - GAS tonsillitis untreated.
- 60% - known rheumatic patients with GAS tonsillitis (recurrence).

Acute rheumatic fever(ARF)

Epidemiology

- **Low-Risk population** - rheumatic fever (ARF) incidence <2 per 100,000 school-age children per year, or all-age rheumatic heart disease (RHD) prevalence of <1 per 1,000 population.
- **Moderate/High-Risk population** - ARF incidence >2 per 100,000 school-age children per year, or all-age RHD prevalence of >1 per 1,000 population.

Etiologic agent

- Group A Beta Hemolytic (GABH) streptococci.
- Gram positive cocci occurring in chains, capsulated with fimbria.
- Rheumatogenic serotypes are:-
1,2,3,5,6,14,12,18,27,24,19,29

Etiology and pathogenesis

- Acute Rheumatic Fever occurs 1-5 weeks after streptococcal throat infection.
- The **cytotoxicity theory** suggests that a GAS toxin is involved
- **Immune-mediated pathogenesis**
Binding of an M-protein N-terminal domain to a region of collagen type IV leads to an **antibody response**

Pathology

- Primarily affects the heart, joints and central nervous system.
- The major importance is its ability to cause fibrosis of heart valves, leading to crippling valvular heart disease, heart failure and death.

Investigations

- Acute phase reactant – ESR, CRP
- Evidence of recent Strep Gp A infection – ASO, throat culture, rapid antigen test, Anti DNAase Ab.
- ECG – prolong PR interval.
- Echocardiography – Valvulitis, Myo/pericarditis, Functional heart assessment.

TABLE II DIAGNOSIS OF RHEUMATIC FEVER

Clinical and laboratory Criteria

Supportive evidence of preceding streptococcal infection (essential except for diagnosis of Chorea)

Major criteria (more specific)

Anti streptolysin O

– Carditis 60-90%

ASO titer: >333 unit for children and > 250 for adults.

– Polyarthrititis 70%

Anti-deoxyribonuclease B

– Chorea 10-30%

(normal values Anti DNase B titer 1:60 unit in preschool, 1:480 units in school children & 1:340 in adults)

– Subcutaneous nodule 0-5%

– Erythema marginatum 0-5%

Minor criteria (less specific)

History of (within previous 45 days)

– Fever

– streptococcal sore throat

– Polyarthralgia

– scarlet fever

– ESR, CRP, polymorphonuclear leukocytosis
(follow standard laboratory values)

– positive throat culture

– positive rapid streptococcal antigen detection test

– *ECG: Prolonged PR interval

* Normal upper range: – PR Interval: 3-12 yrs: 0.16sec; 12-14 y: 0.18 sec; ≥17y: 0.20 sec)

Revised Jones criteria

A. For all patient populations with evidence of preceding GAS infection	
Diagnosis: Initial ARF	2 Major manifestations or 1 major plus 2 minor manifestations
Diagnosis: recurrent ARF	2 Major or 1 major and 2 minor or 3 minor
B. Major criteria	
Low-risk populations*	Moderate- and high-risk populations
Carditis [¶]	Carditis
• Clinical and/or subclinical	• Clinical and/or subclinical
Arthritis	Arthritis
• Polyarthritis only	• Monoarthritis or polyarthritis
	• Polyarthralgia ^Δ
Chorea	Chorea
Erythema marginatum	Erythema marginatum
Subcutaneous nodules	Subcutaneous nodules
C. Minor criteria	
Low-risk populations*	Moderate- and high-risk populations
Polyarthralgia	Monoarthralgia
Fever ($\geq 38.5^{\circ}\text{C}$)	Fever ($\geq 38^{\circ}\text{C}$)
ESR ≥ 60 mm in the first hour and/or CRP ≥ 3.0 mg/dL [◇]	ESR ≥ 30 mm/h and/or CRP ≥ 3.0 mg/dL [◇]
Prolonged PR interval, after accounting for age variability (unless carditis is a major criterion)	Prolonged PR interval, after accounting for age variability (unless carditis is a major criterion)

Evidence of preceding GAS infection is required for both populations.

GAS: group A streptococcal infection; ARF: acute rheumatic fever; ESR: erythrocyte sedimentation rate; CRP: C-reactive protein.

* Low-risk populations are those with ARF incidence ≤ 2 per 100,000 school-aged children or all-age rheumatic heart disease prevalence of ≤ 1 per 1000 population-year.

¶ Subclinical carditis indicates echocardiographic valvulitis.

Δ See section on polyarthralgia, which should only be considered as a major manifestation in moderate- to high-risk populations after exclusion of other causes. As in past versions of the criteria, erythema marginatum and subcutaneous nodules are rarely "stand alone" major criteria. Additionally, joint manifestations can only be considered in either the major or minor categories but not both in the same patient.

◇ CRP value must be greater than upper limit of normal for laboratory. Also, because ESR may evolve during the course of ARF, peak ESR values should be used.

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ARF – Carditis

- Found in 60%-90% of cases, Mainly Valvulitis.
- 30-70% long term morbidity.
- **Mitral Valve** most commonly affected.
- Aortic Valve more specific for diagnosis ARF.
- Acute heart damage is not influenced by treatment given.

ARF – Arthritis

- **2-5 weeks latent period post PSGA infection.**
- **Large joint migratory polyarthrititis.**
- **Rapid response to anti-inflammatory treatment.**
- **No long term morbidity.**

ARF— Sydenham Chorea

- 10–15% of patients with acute RF
- Inflammation involving the basal ganglia, cerebral cortex and cerebellum.
- Diagnostic as single criteria.
- Self limited disease

- **Subcutaneous nodule**
related with significant rheumatic heart disease.



- **ARF-Erythema Marginatum**

Appear in less than 5% of cases.

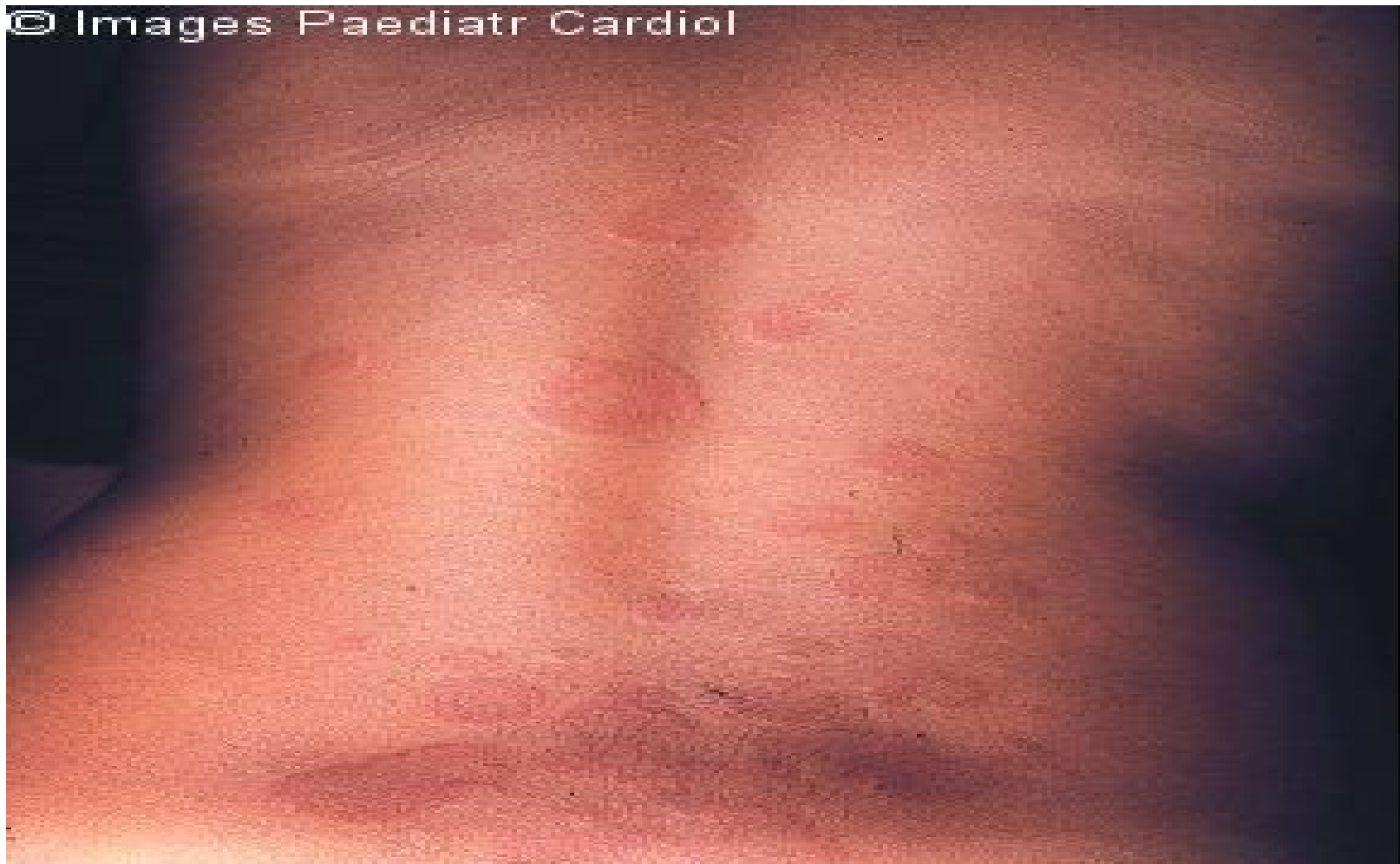
Erythematous, serpiginous[regular border], macular lesions with pale centers .

Occurs primarily on the **trunk and extremities**, but **not on the face**.

Accentuated by **warming the skin**.

Associated with **carditis**

Erythema Marginatum





Jones criteria for the diagnosis of ARF

	Low-risk population ARF incidence ≤ 2 per 100 000 school-aged children or all-age RHD prevalence of ≤ 1 per 1000 population year	Moderate/high-risk population Children not clearly from a low-risk population
Major criteria		
Carditis	Clinical and/or subclinical*	Clinical and/or subclinical*
Arthritis	Polyarthritits	Monoarthritis , polyarthritits and/or polyarthralgia
	Chorea	Chorea
	Erythema marginatum	Erythema marginatum
	Subcutaneous nodules	Subcutaneous nodules
Minor criteria		
Carditis	Prolonged PR interval†	Prolonged PR interval†
Arthralgia	Polyarthralgia	Monoarthralgia
Fever	$\geq 38.5^{\circ}\text{C}$	$\geq 38^{\circ}\text{C}$
Markers of inflammation	Peak ESR ≥ 60 mm in 1 h and/or CRP ≥ 3.0 mg/dL	Peak ESR ≥ 30 mm in 1 h and/or CRP ≥ 3.0 mg/dL

Diagnosis

- **Initial attack** : 2 major manifestations, or 1 major and 2 minor manifestations, plus evidence of recent GAS infection.
- **Recurrent attack** : 2 major, or 1 major and 2 minor, or 3 minor manifestations (the latter only in the Moderate/High-Risk population), plus evidence of recent GAS infection

ARF – Primary Treatment

- **10 days penicillin to eradicate GAS.**
- **High dose Aspirin** (50-100 mg/kg/day) until clinical and laboratory evidence of inflammation resolve.
- **If severe carditis – Steroid (prednisone 2mg/kg/day for 2 weeks and taper down)**

Complications

- Long-term sequelae of RF are essentially limited to the heart
- Approximately 50–70% of patients with carditis recover with no residual heart disease
- The more severe the initial cardiac involvement, the greater the risk for residual heart disease.

ARF : Prophylaxis

- 1° - prevents 1st episode of ARF.
- 2° - prevents recurrences of ARF.
 - Arthritis - at least 5 years
 - Carditis - at least 10 years (recurrence free & no residual heart disease)

Chemoprophylaxis for Recurrences of Acute Rheumatic Fever (Secondary Prophylaxis)

DRUG	DOSE	ROUTE
Penicillin G benzathine	600,000 IU for children weighing ≤ 60 lb and 1.2 million IU for children > 60 lb, every 4 wk*	Intramuscular
or		
Penicillin V	250 mg, twice daily	Oral
or		
Sulfadiazine or sulfisoxazole	0.5 g, once daily for patients weighing ≤ 60 lb	Oral
	1.0 g, once daily for patients weighing > 60 lb	

Secondary prophylaxis

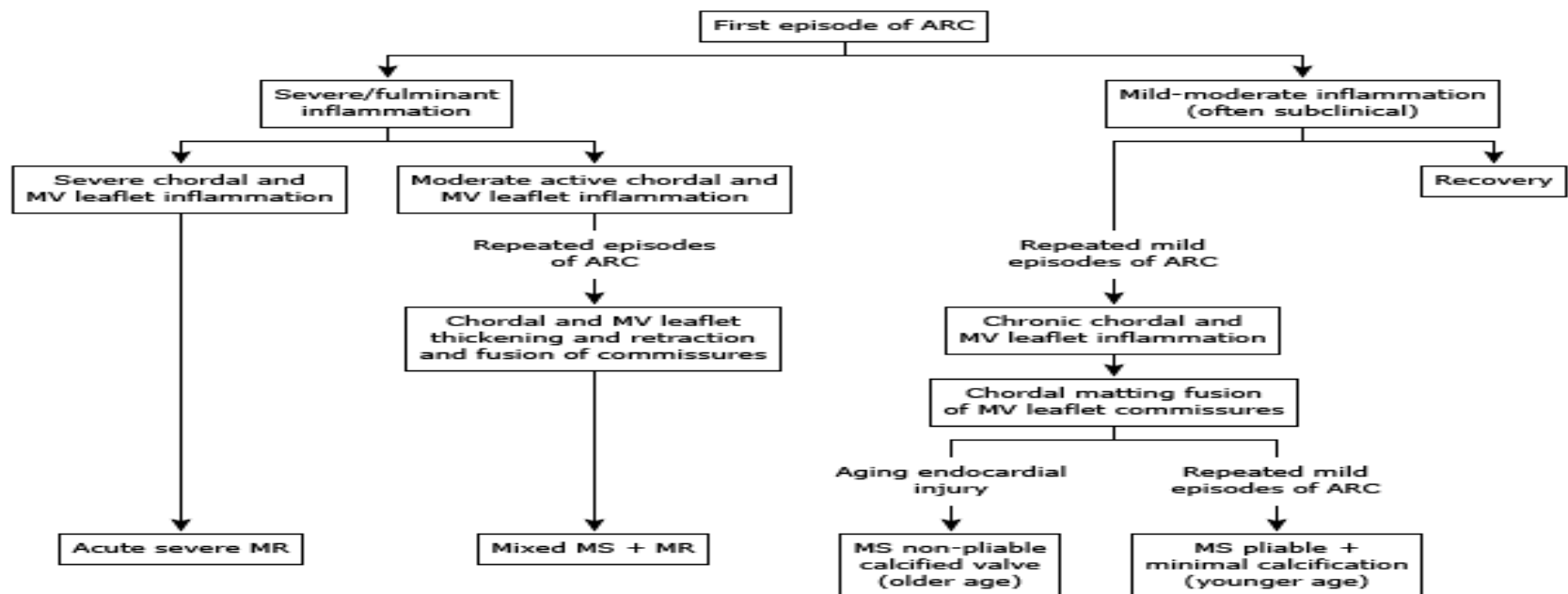
CATEGORY	DURATION
Rheumatic fever without carditis	5 yr or until 21 yr of age, whichever is longer
Rheumatic fever with carditis but without residual heart disease (no valvular disease*)	10 yr or until 21 yr of age, whichever is longer
Rheumatic fever with carditis and residual heart disease (persistent valvular disease*)	10 yr or until 40 yr of age, whichever is longer; sometimes lifelong prophylaxis

Rheumatic heart disease

Introduction

- The valvular lesions begin as small verrucae composed of fibrin and blood cells along the borders of 1 or more of the heart valves.
- The *mitral valve* is affected most often, followed in frequency by the aortic valve.

Pathogenesis of rheumatic mitral valve disease



This figure depicts the theoretical pathway for the various phenotypic manifestations of rheumatic MV disease. The initial inflammatory response following the first episode of ARC determines the degree and extent of structural involvement of the MV apparatus. Severe inflammation of the chordal structures and MV leaflets often lead to severe MR in the very young, whereas moderate inflammation followed by repeated bouts of ARC is more likely to cause mixed MV disease (MR and MS) later in life. Isolated MS is usually the consequence of milder or even subclinical forms of ARC followed by episodes of recurrent carditis. In developing countries where the risk of recurrent ARC is high, MS presents at a younger age and the valve leaflets are more likely to be pliable and not calcified. In Western societies where the risk of recurrent ARC is low, MS presents later in life, and the MV leaflets are more likely to be rigid and calcified.

ARC: acute rheumatic carditis; MS: mitral stenosis; MV: mitral valve; MR: mitral regurgitation.

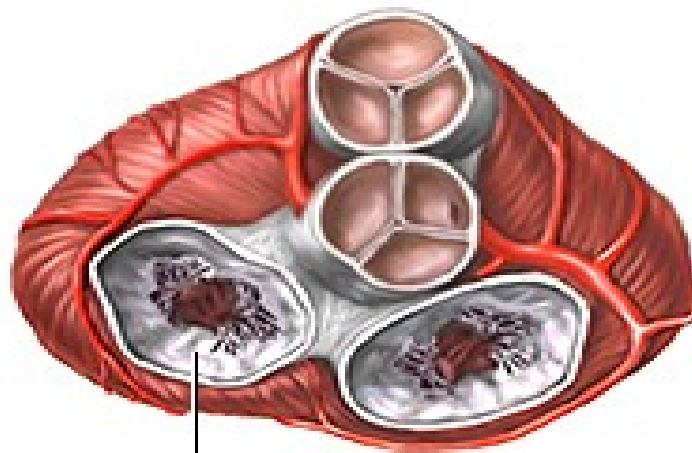
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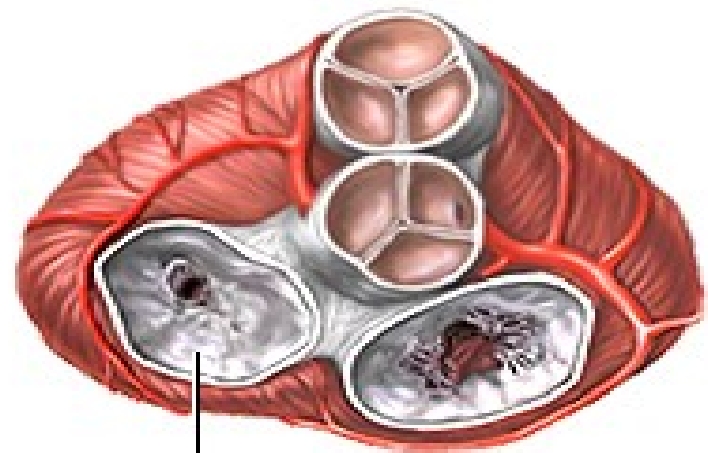
Mitral stenosis

- ❖ Mitral valve is located in the left side of the heart between left atrium and left ventricle.
- ❖ Narrowing of the mitral valve **MVA <2.5 sqcm**
- ❖ Symptom start to occur only when the valve area become **< 2.5cmsq.**

Mitral stenosis



Normal
mitral valve



Narrowing of
mitral valve
(mitral valve stenosis)

Mitral stenosis

Grade	MVA	Pressure gradient	PCWP
• Mild	>1.5	<5mmHg	<10mmHg
• Moderate	1-1.5cm	5-10mmHg	10- 17mmhg
• Severe	<1cm	>10mmHg	>18mmHg

Mitral stenosis causes

- **Rheumatic**, for all practical purpose .

Other theoretical considerations:

- **Congenital**
- **SLE**
- **Rheumatoid Arthritis**
- **Amyloidosis**
- **Malignant Carcinoid**
- **Large vegetation, LA myxoma**

Pathophysiology of Mitral stenosis

- As MV starts narrowing LA pressure $\hat{=}$ PCWP $\hat{=}$ PV pressure $\hat{=}$ interstitial edema.
- Patient may be normal at rest & becomes symptomatic as he/she starts exercising.
- As interstitial edema $\hat{=}$ PV becomes flooded with fluid.

Mitral stenosis

Compensatory mechanisms :

- Lymphatic drainage ↑
- Pulmonary constriction ↑

Over a period time –intimal and muscle hypertrophy and hyperplasia=> fixed pulmonary arterial hypertension.

Mitral stenosis

- History
- Physical examination
- ECG
- Chest X-ray
- Echocardiography
- Cardiac Catheterization

History

- History of Rheumatic Fever, remembered in only **fifty percent of cases**.
- Orthopnea
- Breathlessness on effort-PND
- **Palpitation**
- **Hemoptysis**-usual in children
- Angina(due to severe PHT)
- Dysphagia(compression by esophagus by the enlarged LA)

Physical examination

- **JVP raises** as pulmonary arterial pressure starts rising
- (initially prominent a wave followed by prominent V wave)=>**Hepatomegaly => pedal edema**
- Hepato jugular reflex
- Apex beat normal in its position as the LV is normal
- Tapping apex beat
- **Parasternal heave**(forcible contracting RV)
- **P2 palpable PA** palpable on systolic palpation.

Physical examination

- **Loud S1**
- Opening snap –bell of streetscape- b/n apex & sternal border.
- **Mid diastolic murmur** ==> as severity increases presystolic accentuation can also occur
- As severity increases you can still here loud P2, Grahamsteel murmur
- **Pulmonary ejection murmur heard** in advanced PHT

- The most important to tell the severity of the murmur is not the loudness of the MS murmur but it is the length of the MS murmur, Evidence of CHF and evidence of PHT.

Mitral regurgitation

Chronic mitral regurgitation-

- Gradual onset with gradual development.
- Allows adaptation of the heart
- May not have other features other than palpitation for years.
- LA starts dilating and show increased in compliance
- LV also starts dilating & compensate for forward SV
- Even compensation occur when the patient is doing vigorous exercise.

- Chronic MR doesn't immediately decrease functional state of the patient.
- **It is a chronic processes.**

Causes of chronic MR

- Rheumatic
- MVP(Marfan)
- DCM, RCM, HCM
- Cleft MV
- SLE & RhA
- Prosthetic valve dysfunction
- Infective endocarditis

Diagnosis of MR

- History
- Physical examination
- ECG
- Chest X-ray
- Echocardiography
- Cardiac Catheterization

History

- **Palpitation**(commonest)
- **Breathlessness on effort**
- Orthopnea=>Paroxysmal nocturnal dyspnea (PND)
- Pedal edema
- Acute breathlessness-pulmonary edema=>dramatic deterioration of patient

Physical finding

- Apex beat shift down and out(LV dilation)-
=>Forceful ill sustained (regurgitant lesion)
- Parasternal heave(PAHT)
- Palpable PA,
- Palpable P2
- Raised JVP-> Hepatomegaly => Pedal edema
- Basal creps

Physical finding

- S1 soft
- S3 gallop
- **Pan-systolic murmur** best heard at the apex radiating to **Lt. axilla.**
- All features of PAHT:-
 - **Loud P2-Grhamsteel**
 - **Mid systolic murmur**
 - **Pulmonary ejection murmur**

Management principle

1. **Treat acute complications** :

Arthritis, CHF, Sydenham chorea, Infective endocarditis, Atrial fibrillation, pulmonary hypertension, Cardiogenic shock etc.

2. Surgery-valve repair or replacement.

3. Anticoagulation

4. Secondary prophylaxis[prevention of recurrent

Infective Endocarditis

Infective endocarditis

- Definition: infection of the **endocardium** and/or **heart valves, chordae tendinae** or **sites of shunting septal defects**.
- IE is not a uniform and varies with
 - ✓ Clinical manifestation
 - ✓ Underlying cardiac illness
 - ✓ Micro-organism involved

Microbiology

Most cases are caused by relatively small number of micro-organisms.

Gram positive cocci account for **90%** of cases (adult studies).

Streptococcus viridans – are the most common causes at **all ages**.

Staphylococcus aureus & CONS – 2nd most common organisms.

- About 75-90% of cases after the neonatal period are associated with an **underlying cardiac diseases**
- **Neonatal and infantile** IE is rare & in most cases **occur in structurally normal hearts .**
- ❖ **Poor dental hygiene results in a greater risk for endocarditis after brushing teeth & chewing hard food.**

Pathogenesis

Modes of endothelial injury

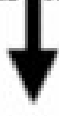
- High velocity jet
- Flow from high pressure to low pressure chamber
- Flow across narrow orifice of high velocity
- Bacteria deposited on edges of low pressure sink or site of jet impaction

- Pressure gradient required producing a high-velocity jet stream
- The high flow results in a Venture effect (a low pressure area adjacent to the area of high flow)
- **Bacteria to settle in this area of low pressure.**

Valvular Endothelium



Platelet-Fibrin Deposition



**Nonbacterial Thrombotic
Endocarditis**



Mucous Membranes



Trauma



Bacteremia



Adherence



Colonization



Vegetation = Hallmark

Clinical Feature

- **Fever** is present in almost all cases.

- **Anemia**

- ❖ Cardiac Manifestations: **sign& symptoms of underlying cardiac lesion and CHF (33%)**

- **New or changing heart murmurs** are common.

Conduction block, **valvular incompetence** .

- ❖ **Embolic Manifestation**

- ❖ **Immune complex deposit**

Splenomegaly(70)%

Laboratory Diagnosis

- Blood culture **at least three bottle** from separate venipuncture continuous
- **CBC,ESR,Urine analysis**

Diagnostic Criteria

Duke criteria

There **are three** categories

Modified Duke criteria for diagnosis of infective endocarditis: Table A

Definite IE is established in the presence of any of the following:
Pathologic criteria
Pathologic lesions – Vegetation or intracardiac abscess demonstrating active endocarditis on histology, OR
Microorganism – Demonstrated by culture or histology of a vegetation or intracardiac abscess
Clinical criteria
Using specific definitions listed in Table B:
2 major clinical criteria, OR
1 major and 3 minor clinical criteria, OR
5 minor clinical criteria
Possible IE*
Presence of 1 major and 1 minor clinical criteria OR presence of 3 minor clinical criteria
Rejected IE
A firm alternate diagnosis is made, OR
Resolution of clinical manifestations occurs after ≤ 4 days of antibiotic therapy, OR
No pathologic evidence of infective endocarditis is found at surgery or autopsy after antibiotic therapy for 4 days or less
Clinical criteria for possible or definite IE not met

IE: infective endocarditis.

* The category of possible IE represents a modification from the previous published Duke criteria.

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Modified Duke criteria for diagnosis of infective endocarditis: Table B

Major criteria
Positive blood cultures for IE (1 of the following):
Typical microorganisms consistent with IE from 2 separate blood cultures:
<i>Staphylococcus aureus</i>
Viridans streptococci
<i>Streptococcus gallolyticus</i> (formerly <i>S. bovis</i>), including nutritional variant strains (<i>Granulicatella</i> spp and <i>Abiotrophia defectiva</i>)
HACEK group – <i>Haemophilus aphrophilus</i> (subsequently called <i>Aggregatibacter aphrophilus</i> and <i>Aggregatibacter paraphrophilus</i>), <i>Actinobacillus actinomycetemcomitans</i> (subsequently called <i>Aggregatibacter actinomycetemcomitans</i>), <i>Cardiobacterium hominis</i> , <i>Eikenella corrodens</i> , <i>Kingella kingae</i>
Community-acquired enterococci in the absence of a primary focus, OR
Persistently positive blood culture:
For organisms that are typical causes of IE – At least 2 positive blood cultures from blood samples drawn >12 hours apart
For organisms that are more commonly skin contaminants – 3 or a majority of ≥4 separate blood cultures (with first and last drawn at least 1 hour apart)
Single positive blood culture for <i>Coxiella burnetii</i> or phase I IgG antibody titer >1:800*
Evidence of endocardial involvement (1 of the following):
Echocardiogram positive for IE:
Vegetation (oscillating intracardiac mass on a valve or on supporting structures, in the path of regurgitant jets, or on implanted material, in the absence of an alternative anatomic explanation), OR
Abscess, OR
New partial dehiscence of prosthetic valve
New valvular regurgitation
Increase in or change in preexisting murmur not sufficient
Minor criteria
Predisposition – Intravenous drug use or presence of a predisposing heart condition (prosthetic heart valve or a valve lesion associated with significant regurgitation or turbulence of blood flow)
Fever – Temperature ≥38.0°C (100.4°F)
Vascular phenomena – Major arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial hemorrhage, conjunctival hemorrhages, or Janeway lesions
Immunologic phenomena – Glomerulonephritis, Osler nodes, Roth spots, or rheumatoid factor
Microbiologic evidence – Positive blood cultures that do not meet major criteria, OR serologic evidence of active infection with organism consistent with IE
(Echocardiographic minor criteria eliminated)*

IE: infective endocarditis; IgG: immunoglobulin G.

* Modifications from the previous published Duke criteria are noted by the asterisk.

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Cont..

Clinical criteria: 1) definite

- 2 major criteria; or
- 1 major criterion and 3 minor criteria; or
- 5 minor criteria

Possible IE:

- 1 major criterion and 1 minor criterion; or
- 3 minor criteria

Cont..

Rejected IE:

- Firm alternative diagnosis explaining evidence of IE;
or
- Resolution of IE syndrome with antibiotic therapy
within 4 days; or
- No pathologic evidence of IE at surgery or autopsy,
with antibiotic with for 4 days
- Does not meet criteria for possible IE as above

management

Principles

Eradication of causative organism

Rx of heart failure

Surgical management.

Management of complication

Prevention

Cont...

Eradication of causative organism

Antibiotic shall be

- 1) **high dose with high serum conc.**
- 2) **bactericidal**
- 3) **in combination with synergistic abc.**
- 4) **prolonged period Rx**
- 5) **often parenteral**

WHY?

1) vegetation & biofilm is avascular, so abc reach only by diffusion

2) high colony of bact. in vege

3) low multiplication rate of bacteria in vegetation

The initial antibiotic should target

NVE and late PVE

❖ In general ***Ampicillin-sulbactam and Gentamicin are choice of antibiotics.***

Patients with prior histories of ***penicillin allergy*** can usually be treated with ***ceftriaxone***

Duration of therapy

➤ Depends on identified organism and nature of valve.

1) NVE caused by *S.viridans* 2-4wks.

2) NVE caused by others 4-6wks

3) All PVE treated for at least 6wks.

Indication for surgery.

- Anterior mitral leaflet vegetation, with size >10 mm.
- One or more embolic event during 1st 2wk antibiotic
- Valvular dysfunction.
- HF unresponsive to medical therapy.
- New heart block , Perivalvular extension
- Failure to sterilize the blood despite adequate antibiotic levels.

Complication of IE and prognosis

- Cardiac: Myocardial abscesses and toxic myocarditis
- Life-threatening arrhythmias & aneurism.
- Systemic emboli : CNS & renal manifestations.
- Pulmonary emboli may occur in children with VSD or TOF,
- Prognosis: despite the use of antibiotic , mortality is at 20-25%.
- Poor prognostic factor : PVE, presence of complication, S.aureus fungal infection,some Echo finding.

Prevention of IE

- Antimicrobial prophylaxis before various procedures
- Proper general dental care and oral hygiene are most important in decreasing the risk of IE
- Vigorous treatment of sepsis and local infections decrease risk of IE
- Parenteral antibiotics recommended in high-risk patients

Acquired heart disease in children II

Acute Myocarditis

Definition-: an inflammatory infiltration of the myocardium with associated necrosis, degeneration or both.

- WHO (marburg Criteria.1996)- A minimum of 14 infiltrating leucocytes per mm, preferably T lymphocytes and up to 4 macrophages may be included.

Contd...

- The incidence and prevalence not known because of large numbers of asymptomatic cases.
- The severe forms are observed predominantly in the age below 3 years.
- The mild forms are seen predominantly in preschool and school age children.

Etiology

Infections:

- Viruses -adeno virus, coxsackievirus, echovirus, and many other viruses including Human immunodeficiency viruses.
- Rarely- bacteria, rickettsia, fungi, protozoa, and parasites.
- HIV-1 virion.

Etiology

Non infectious

- A cell mediated immunologic reaction.
- Cardio toxic drugs-cyclophosphamide, anthracyclins
- Connective tissue disorders(SLE, RA)
- Acute rheumatic fever

Etiology cont..

- Immune-mediated disease:- Acute fever, Kawasaki disease
- Collagen vascular diseases:- SLE, ARF
- Toxic myocarditis: drug ingestion, diphtheria exotoxin & anoxic agents.

Clinical manifestation

- History of upper respiratory tract infection may precede presentation in older children.
- Fever, shortness of breath; headache,
- Pale skin, weakness, decline of appetite

Clinical manifestations

- Chest pain due to coexisting pericarditis.
- Enlarged heart, Tachycardia, Cardiac dysfunction.
- Soft systolic murmur on the apex,
- Decreased arterial pressure.

Physical Examination (Myocarditis)

- Wheezing on lung auscultation,
- enlargement of liver,
- Edema of the extremities,

Laboratory workup

- CBC
- ↑ acute phase reactant
- Serum viral antibody titer
- ANA, anti-Ro/SSA
- Serum cardiac enzymes:-CPK, Troponin I, T
- ECG
- Echocardiogram
- Gallium scanning
- Gadollinium-enhanced MRI
- Antimyosin scintigraphy

Laboratory investigations

- Creatine phosphokinase(CPK) elevated in 7.5% cases,
- Troponin I/T elevated in 50% cases.
- Brain natriuretic peptide(BNP)

Electrocardiography

- Decrease of waves voltage
- Conduction impairments
- Arrhythmias-Atrial Fibrillation

ECG :tachycardia, reduced QRS complex voltage,
ST-segment and T-wave abnormalities

**Premature ventricular
beats**



**Paroxysmal
tachycardia**



Echocardiography

- ☐ Myocardium is diffusely thick,
- ☐ Dilatation of chambers,
- ☐ Reduced myocardial contractility,
- ☐ Possible pericardial effusion.

Management of Acute myocarditis

- i. Heart failure management
- ii. Arrhythmia management
- iii. Exercise restriction
- iv. Follow up-because persistent chronic inflammation
- v. May lead to dilated cardiomyopathy

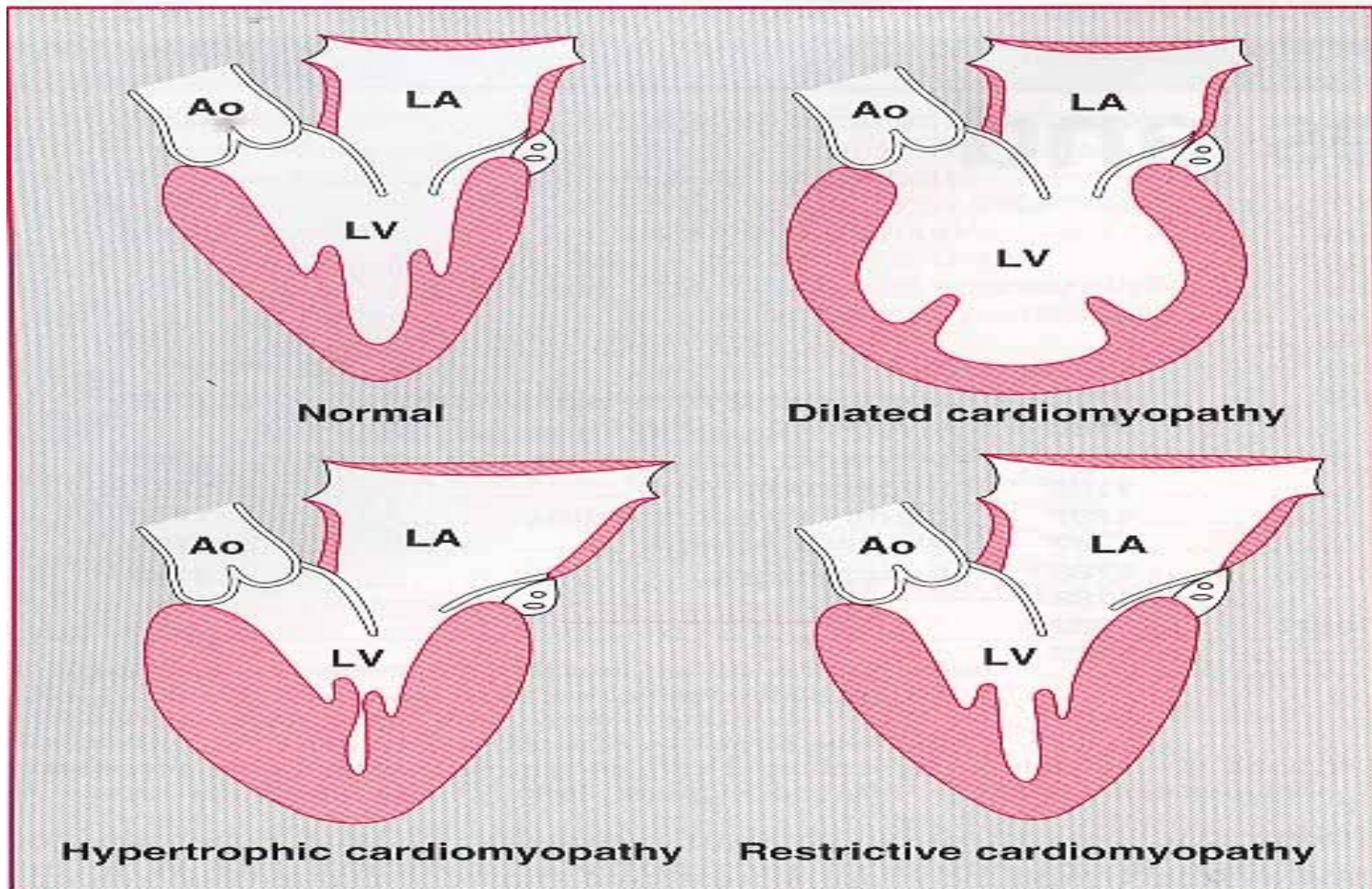
Cardiomyopathies

Cardiomyopathies

- There are several types of cardiomyopathies defined by the WHO:
- “Dilated cardiomyopathy” or DCM,
- “Hypertrophic cardiomyopathy” or HCM,
- “Restrictive cardiomyopathy” or RCM,

Other-Arrhythmogenic Right Ventricular
Dysplasia (ARVD)

Cardiomyopathies



Incidence & Prognosis cardio..

- 3-10 cases per 100,000 population.
- 1% of all pediatric cardiac diseases.
- During infancy incidence is 10 times higher in than older children.

Incidence & Prognosis cardio..

- Death from progressive pump failure

1-year	25%
2-year	35-40%
5-year	40-80%
- Stabilization observed in 20-50% of patient.
- Complete recovery is rare.

Classification

- Primary

- ❖ genetic abnormalities of cardiac muscle)

- Dilated
- Hypertrophic
- Restrictive

- Secondary

infections, metabolic and nutritional diseases, endocrine disorders, neuromuscular diseases, blood diseases, tumors)

Dilated Cardiomyopathy

- A progressive disease of heart muscle that is characterized by LV dilatation & systolic dysfunction.

Dilated Cardiomyopathy

—Pathology:- Increased heart size & weight.

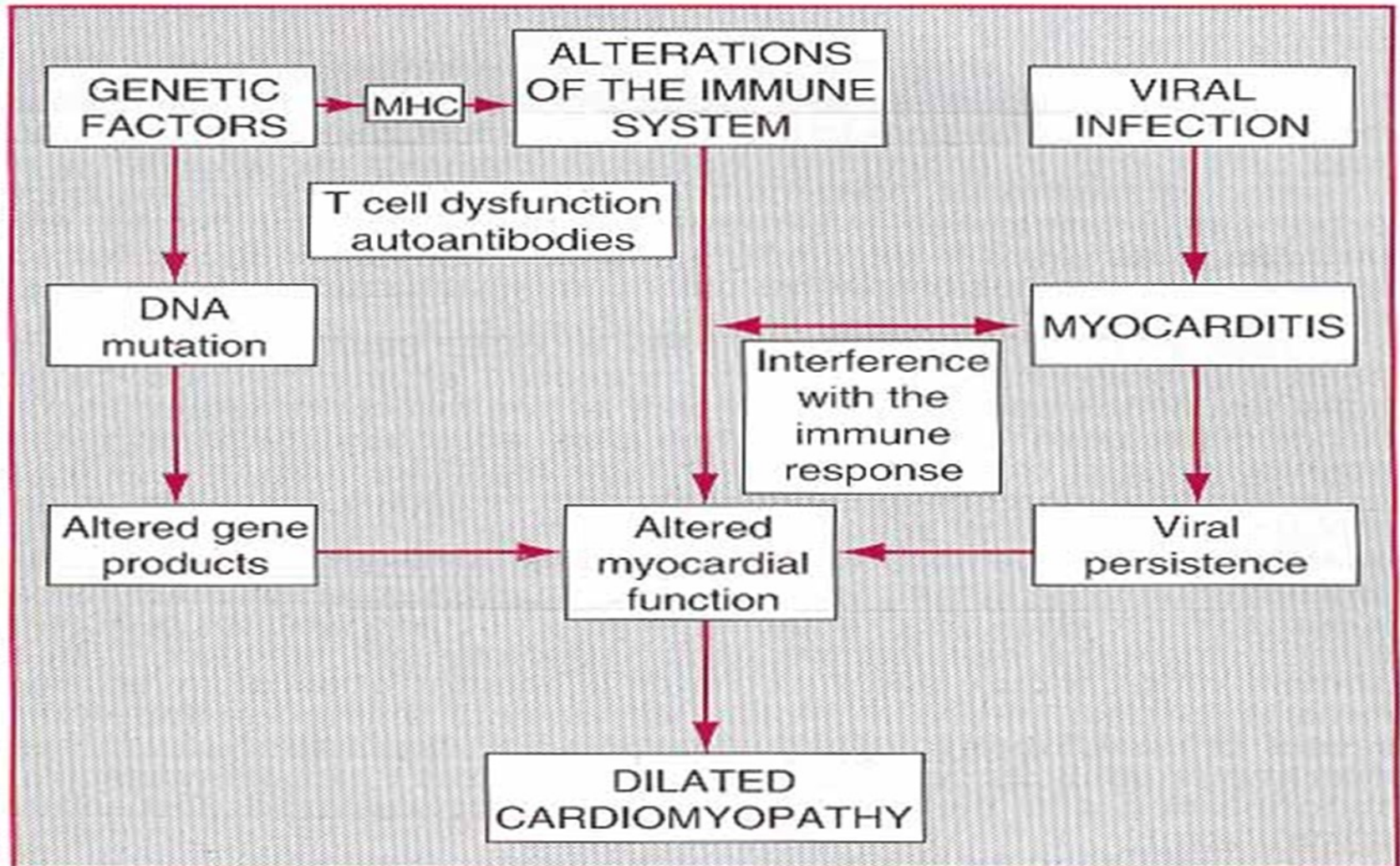
- Normal wall thickness.
- Heart dysfunction out of portion to fibrosis.

Dilated Cardiomyopathy

Etiology:-

- Genetic basis or
- Sequelae of **viral myocarditis**.
- **In 20-50 % familial(Autosomal dominant inheritance is most commonly encountered)**
- Mutations in several cardiac structural or metabolic genes have been identified.

Dilated CMP-Pathophysiology



Dilated Cardiomyopathy

LV Failure=>  increase in LVEDV =>
 PV and  PCWP.

❖ MV regurgitation result from papillary muscle dysfunction or severe dilatation of mitral valve annulus.

Clinical Manifestations

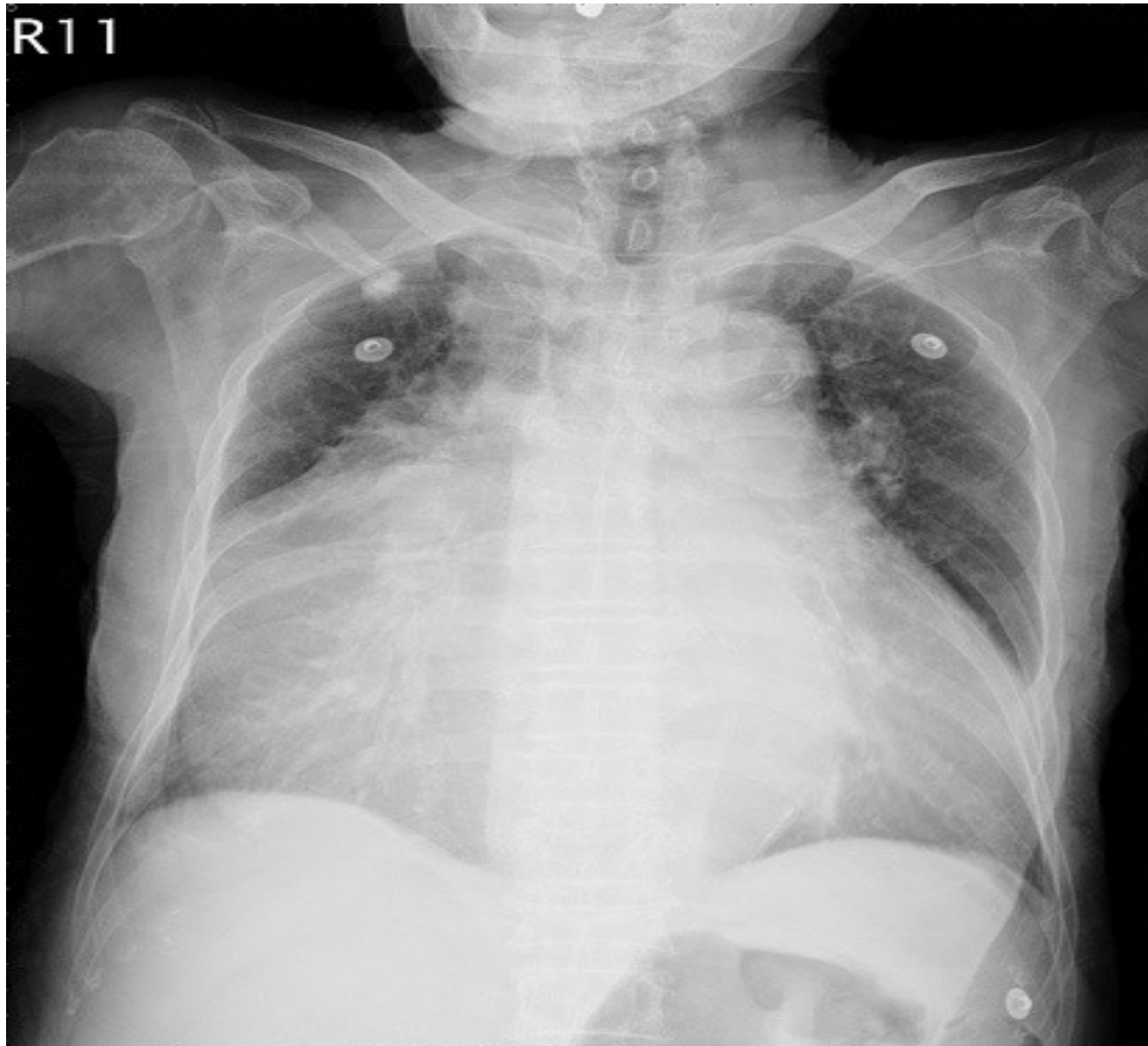
- Symptoms of heart failure occur insidiously, but some time suddenly;
- Irritability, anorexia, abdominal pain,
- Cough, dyspnea with exertion from pulmonary congestion.
- Respiratory symptoms and failure to thrive in Infants and younger children.

Clinical M/S cont..

- Cool and pale skin ;
- Decreased arterial pulse , tachycardia, palpitation.
- Increased Jugular venous pressure , hepatomegaly, edema;
- Holosystolic murmurs of mitral and tricuspid insufficiency;
- Blood pressure may be low,& pulse pressure may be narrow;
- S3 gallop rhythm is audible.
- Cardiomegaly

Diagnosis

- Clinical signs
- Chest x-ray
- CT Scan
- Echocardiogram
- 12-lead ECG
- Heart biopsy
- MRI (magnetic resonance imaging)
- Cardiac catheterization



ECG

- Atrial enlargement, varying degrees of left or right ventricular hypertrophy;

Doppler studies

Echocardiography

1. Decreased flow velocity through the aortic valve.
2. Mitral regurgitation.
3. In long-standing cases, evidence of Pulmonary hypertension may exist.

Serious complications

- ✓ Ventricular arrhythmias => syncope and sudden death,
 - ✓ Pulmonary or systemic emboli from intra-cardiac thrombi.
 - ✓ Development of PVD from chronically left atrial pressure.
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Prognosis

- Progressively down hill, although some patients may remain stable for years.
- ❖ Treatment of HF may result in temporary remission,
- ❖ Relapses are common and in time – patients become resistant to therapy.
- Prognosis for survival beyond a year is poor.

Management

- Limit activity.
- Salt restriction.
- Fluid restriction for significant low Na⁺.
- Initiate medical therapy.
- ACE inhibitors,
- Loop diuretics.
- Digoxin,
- Beta –blockers (metoprolol, carvedilol).
- Hydralazine/nitrate combination.

- Anticoagulation (If there is hx of thromboembolism, presence of mural thrombi).
- Intravenous dopamine / dobutamine in case of cardiogenic shock.
- Cardiac transplantation.

Acute Pericarditis

- Causes may be idiopathic or
- Secondary causes includes
 - Viruses (coxsackie, flu, EBV, mumps)
 - Bacteria (pneumonia, ARF, TB, Staph, Streps,
 - Fungi
- Drugs: Penicillin, INH, Hydralazine, Cancer drugs
- Others: uremia, RA, SLE, myxedema, malignancy, sarcoidosis

Acute Pericarditis

- ✓ usually a benign course
- ✓ virulent bacteria (H. flu, E. coli) can cause constrictive pericarditis and subsequent tamponade
- ✓ uncomplicated pericarditis usually responds to rest and anti-inflammatories

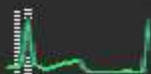
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Clinical Features-pericarditis

- Chest pain
 - central
 - worse on inspiration and lying flat
 - Relief with sitting forward
 - Pericardial friction rub
 - Signs of pericardial effusion, cardiactamponade
 - Fever may be there

Investigating-pericarditis

ECG

- Saddle shaped ST segment elevation but may be normal also

Blood test

- CBC,ESR
- Cardiac enzymes(troponine may be elevated)
- Blood culture,autoantibodies,viral serology, thyroid function test

CXR (Cardiomegally,pericardial effussion)

ECHO (pericardial effussion)

Treatment

Bed rest and oral non steroidal anti-inflammatory drugs (NSAIDs).

If NSAIDs are unsuccessful, steroids may be considered once bacterial causes have been excluded.

Colchicine has not been well studied in the pediatric population

Bacterial pericarditis should be treated for at least 3 to 4 weeks with intravenous antibiotics.

Complete resolution usually in <6 weeks.

Relapse occurs in a small subset of patients, typically improving with reinstitution of NSAIDs or steroids.

Constrictive pericarditis rarely occurs as a late complication of viral pericarditis.

Prognosis

- Survival for patients with bacterial pericarditis currently is >90%
- Risk factors for poor outcome include young age at diagnosis, septicemia, tamponade, delay in diagnosis, inadequate drainage, concurrent myocarditis, and a staphylococcal etiology
- Constrictive pericarditis can be a late complication , most commonly associated with *Staph.aureus*, *H.influenzae*, or *S.pneumoniae* infections.

Tuberculous Pericarditis

- The typical onset is insidious with symptoms of low-grade fever, night sweats, weight loss, malaise, dyspnea, and chest pain.
- The presentation may be complicated by subacute pericardial tamponade.
- Direct extension of miliary tuberculosis or lymphatic spread into the pericardium.
- Hematogenous spread may occur without evidence of pulmonary infiltrates.
- Most patients have a positive Mantoux skin test.

- The pericardial fluid, when aspirated, is sero sanguineous or hemorrhagic with lymphocyte predominance.
- Pericardial biopsy is useful to provide histologic confirmation.
- Pericardial fluid adenosine deaminase levels are diagnostic for tuberculous pericarditis if >50 U/L.
- Mycobacterium cultures may take up to 6 weeks to grow; thus, treatment should be initiated before confirmation of diagnosis.

- ❖ Steroid therapy may be useful to reduce inflammation and increase pericardial fluid resorption .
- ❖ Early in the course of the disease, pericardiectomy is difficult due to the presence of diffuse inflammatory and caseous material.
- ❖ Constrictive pericarditis may develop after recovery.
- ❖ Worldwide, tuberculous pericarditis is one of the leading causes of constrictive pericarditis .

HIV and Other Infections

- Pericardial effusions are common in patients with human immunodeficiency virus (HIV) and are present in up to 25% of children with HIV infection .
- Children with HIV rarely develop tamponade. Pericardial fluid cultures typically are negative.

Thank You